

NETWORKING COMPREHENSIVE REVISION GUIDE

Full 7-Chapter Guide for Last-Minute Revision

CHAPTER 1: INTRO TO NETWORKS

Exam Def: Collection of 2+ devices connected to share data, hardware (printers), and internet.

Trick: Network = Connect + Communicate + Share

Types of Networks

- **PAN:** Personal (1-10m) - Bluetooth.
- **LAN:** Local (Building) - School/Lab.
- **MAN:** Metro (City) - Cable TV.
- **WAN:** Wide (World) - Internet.

Network Topologies

- **Bus:** Single main backbone cable. Cheap.
- **Star:** Central Hub/Switch. Easiest to maintain.
- **Ring:** Circle path. Data flows one way.
- **Mesh:** Reliable, point-to-point connections.
- **Tree:** Hierarchical (Star + Bus combo).

OSI & TCP/IP Models

Layer	Unit	Device
App/Pres/Sess	Data	-
Transport	Segment	Gateways
Network	Packet	Router
Data Link	Frame	Switch/Bridge
Physical	Bits	Hub/Repeater

Mnemonic: All People Seem To Need Data Processing.

CHAPTER 2: PHYSICAL LAYER

L1 Task: Transmitting raw bits through hardware.

Media & Signals

- **Twisted Pair:** Copper, Telephone.
- **Coaxial:** Broadband/TV Cable.
- **Optical Fiber:** Fast Light signals.
- **Unguided:** Radio, Microwaves, Infrared.

NRZ: Binary to Signal without middle transition.

Manchester: Change at the middle of every bit.

Trick: NRZ = Simple; Manchester = Sync.

CHAPTER 3: DATA LINK LAYER

L2 Task: Node-to-node delivery, Framing, Error detection.

Flow & Error Control

- **Stop & Wait:** Send 1, Wait for ACK.
- **Sliding Window:** Send multiple, higher speed.
- **Parity:** Simple bit count.
- **CRC:** Polynomial division (Very Accurate).
- **Checksum:** Mathematical addition check.

Standards (IEEE)

- **802.3:** Ethernet (Wired LAN)
- **802.4:** Token Bus
- **802.5:** Token Ring
- **ALOHA:** Pure (Anytime) vs Slotted (Slots).

CHAPTER 4: NETWORK LAYER

L3 Task: IP addressing and Routing Packets.

IP Addressing

- **IPv4:** 32-bit. Class A (1-126), B (128-191), C (192-223), D (Multicast).
- **IPv6:** 128-bit. Hexadecimal format.

Routing Protocols

- **RIP:** Distance Vector (Max 15 Hops).
- **OSPF:** Link State (Cost-based).
- **BGP:** Internet backbone routing.
- **NAT:** Converts Private IP to Public IP.
- **ICMP:** Error reporting / Ping tool.

NETWORKING GUIDE (PART 2)

CHAPTER 5: TRANSPORT LAYER

L4 Task: End-to-end reliability and port addressing.

TCP vs UDP

- **TCP:** Connection-oriented (SYN, SYN-ACK, ACK), Reliable, Slow.
- **UDP:** Connectionless, Unreliable, Fast (Streaming/Games).

Congestion Control

- **Leaky Bucket:** Shapes traffic into a steady flow. Drops excess.
- **Token Bucket:** Allows bursts of traffic if tokens are available.

Critical Port Numbers

Service	Port	Protocol
HTTP	80	Web
HTTPS	443	Secure Web
FTP	21	Files
SMTP	25	Email Sent
DNS	53	Names

CHAPTER 6: APPLICATION LAYER

L7 Task: Interface between user and network.

Core Services

- **DNS:** Converts Domain Names to IP Addresses.
- **POP3:** Downloads mail and deletes from server.
- **IMAP:** Keeps mail on server (Synchronization).
- **SSH:** Secure/Encrypted Remote Login (Port 22).
- **Telnet:** Insecure remote login.
- **SNMP:** Used for managing network devices.

CHAPTER 7: EMERGING CONCEPTS

Cloud & Internet of Things (IoT)

- **Cloud:** Remote storage and computing.
- **IoT:** Connecting smart devices (Bulbs, ACs).

IoT Communication Protocols

- **MQTT:** Pub-Sub model with a Broker (Lightweight).
- **CoAP:** Client-Server model (Request-Response).

Software Defined Networking (SDN)

Moving the "brain" (Control Plane) to a central controller, leaving routers as "workers" (Data Plane).

Memory Trick: SDN Controller = Principal; Routers = Students.

Final Tip: Focus on Layer Functions and Mnemonics for identifying protocols. Good Luck!